

CV FACE AUTHENTICATION AND DETECTION

Problem Statement ID – 1683

Problem Statement Title - Development of AI/ML based solution for detection of face-swap based deep fake videos

Theme – Miscellaneous

Problem Statement Category – Software

Team ID – 27815

Team Name – CodeCreators

About this Model

In this model Creatively uses OpenCV for real-time deep fake detection by applying frame-by-frame analysis and template matching, making it a lightweight, accessible solution. It's both innovative and flexible, allowing for future integration of AI/ML models to enhance detection capabilities.

PROPOSED SOLUTION & INNOVATION

1. **Real-time Detection:** Enables efficient and fast processing of deep fake videos in real-time environments.



2. **Feature-Based Analysis:** Utilizes facial landmarks and texture analysis for accurate detection without heavy computational demands.



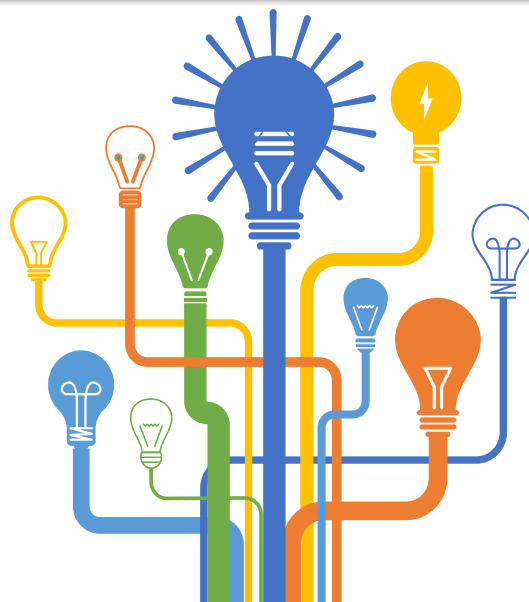
4. **Community-Driven Development:** Leverages OpenCV's open-source ecosystem for continuous innovation and improvement.



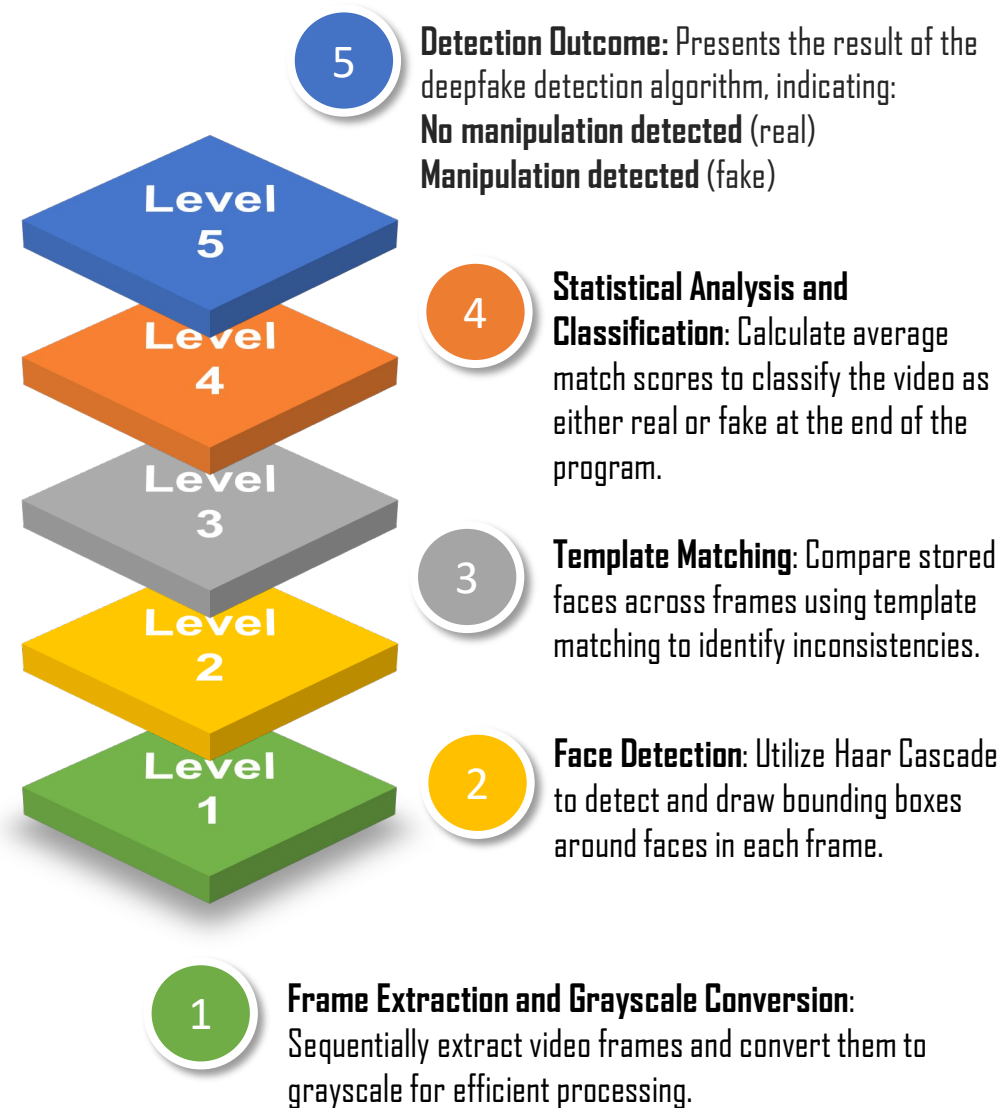
3. **Scalable Deployment:** Easily adaptable for various platforms, including mobile and embedded systems.



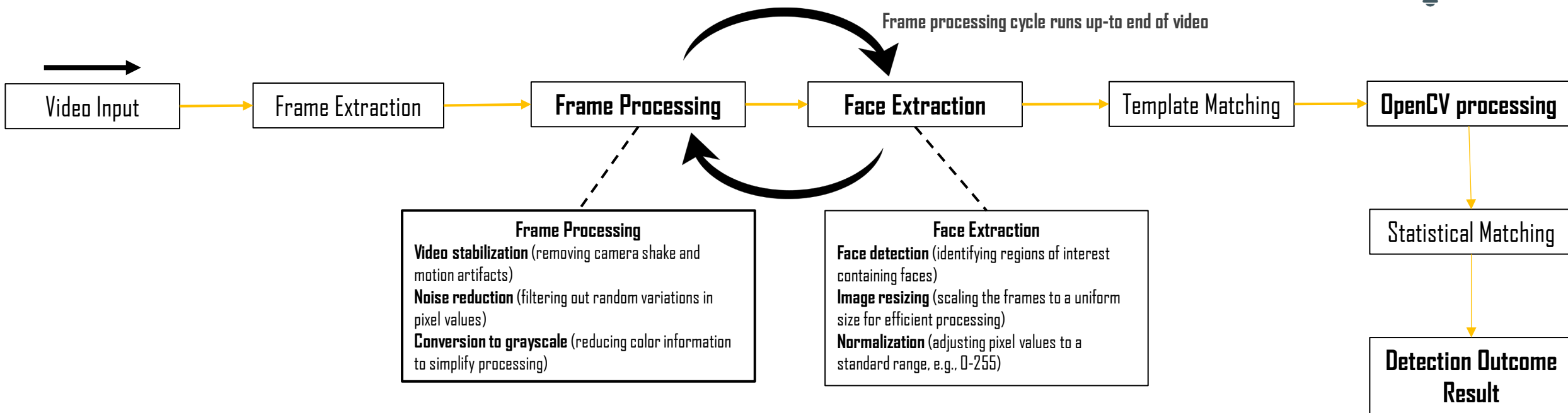
5. **Multi-Stage Filtering:** Reduces false positives by refining detections through sequential validation steps.



METHODOLOGY



CODE PROCESS OF IMPLEMENTATION



TECHNOLOGIES & CONFIGURATIONS FOR DEEP FAKE DETECTION

 **python™** The primary language for implementing the deep fake detection algorithm, leveraging its robust libraries like OpenCV for image and video processing.

Frameworks & Libraries

 **PyTorch**

(optional)

 **NumPy**

 **TensorFlow**

(optional)



Special Features in Open CV

- Open Source
- Cross Platform
- Real Time
- Deep Learning
- Fast Processing
- 3D Reconstruction
- Large Community
- Regular Updates
- Image Filtering
- Object Tracking

Hardware & Configuration

SYSTEM

CPU: i7/i9 Intel or Ryzen 7/9

RAM: 16-32GB

Storage: 512GB-1TB SSD

OS: Win 10/11 (64-bit) or Ubuntu 20.04 LTS+

GPU

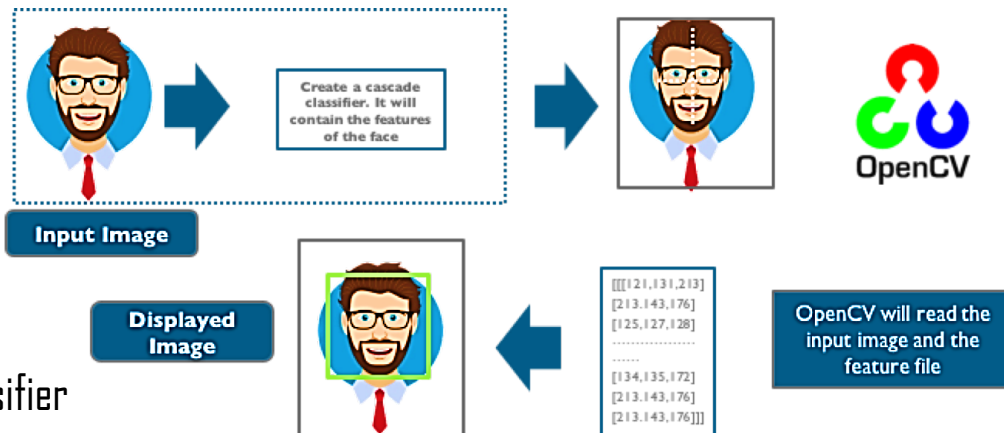
NVIDIA GeForce RTX 3070/80/90 or 40 Series

8GB GDDR6 VRAM

Detection Outcome: Presents the result of the deepfake detection algorithm, indicating:

- No manipulation detected (real)**
- Manipulation detected (fake)**

This script appears to be a face detection and comparison tool using OpenCV. Here's a breakdown of its feasibility and viability



Haar Cascade Classifier

Feasibility Factors:

1. Technical Feasibility:

- Face detection: 8/10 (OpenCV provides robust face detection algorithms)
- Face comparison: 6/10 (template matching has limitations, more advanced methods may be needed)

2. Resource Feasibility:

- Computational resources: 7/10 (may require significant resources for large videos or high-resolution frames)
- Storage resources: 8/10 (face data and video storage requirements are manageable)

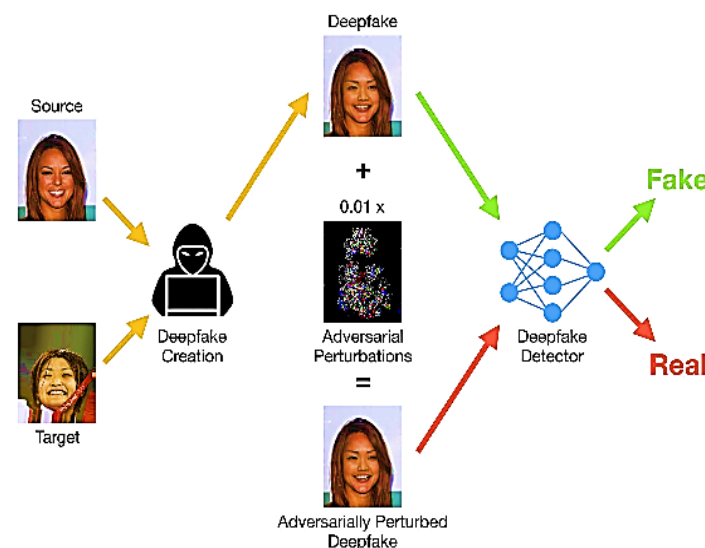
Viability:

1. Performance:

The script may become computationally expensive for large videos or high-resolution frames, as it processes each frame and compares faces in real-time.

2. Accuracy:

The face comparison method used may not be robust enough for varying lighting conditions, angles, or facial expressions.



Potential Risks & Harms

Psychological Harm

- Sextortion
- Defamation
- Undermining trust

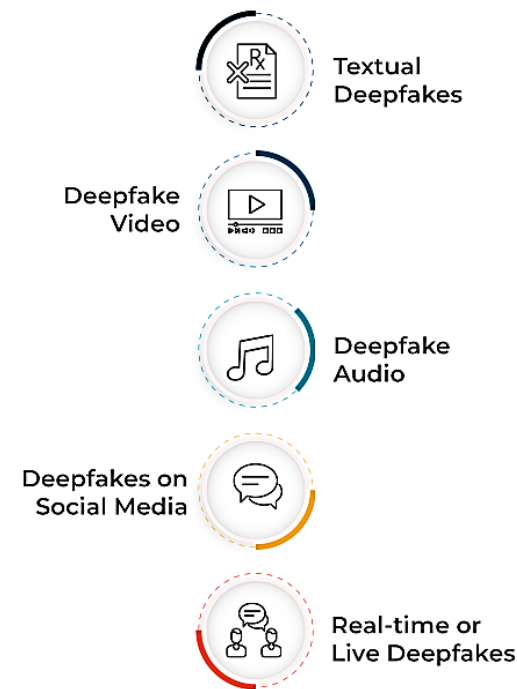
Financial Harm

- Extortion
- Identity theft
- Reputational Damage

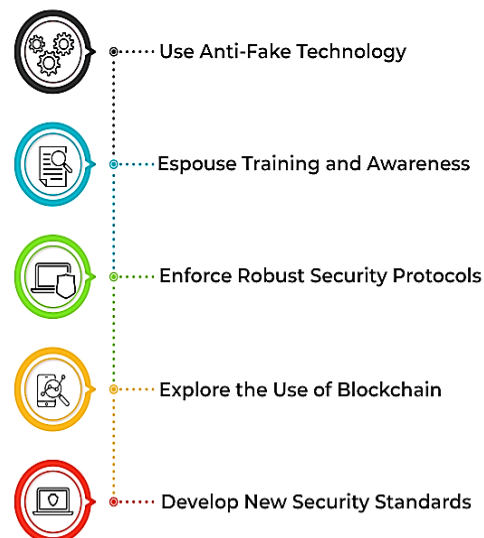
Societal Harm:

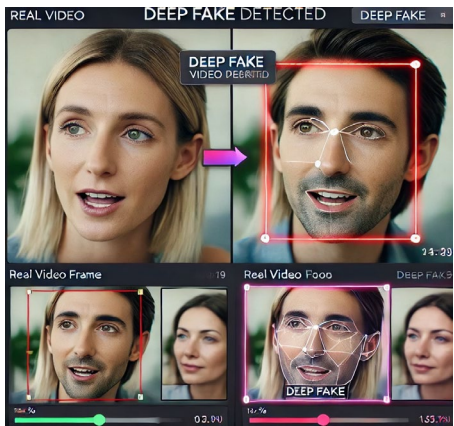
- News media manipulation
- Damage to democracy
- Damage to economic stability

TYPES OF DEEPPAKE FRAUDS



DEEPPAKE FRAUD PREVENTION BEST PRACTICES

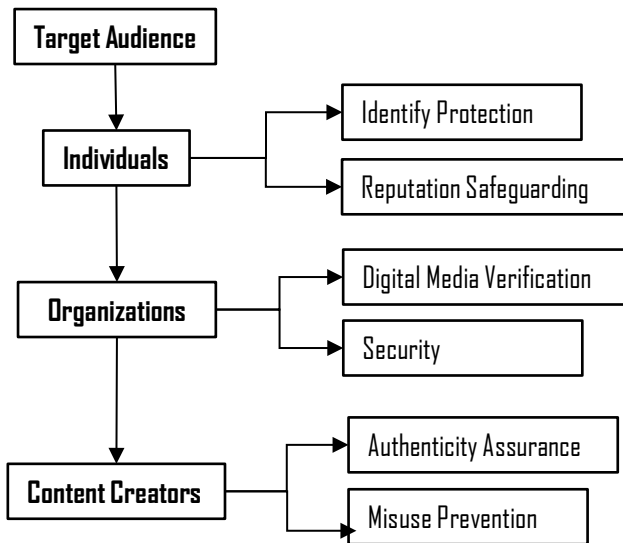
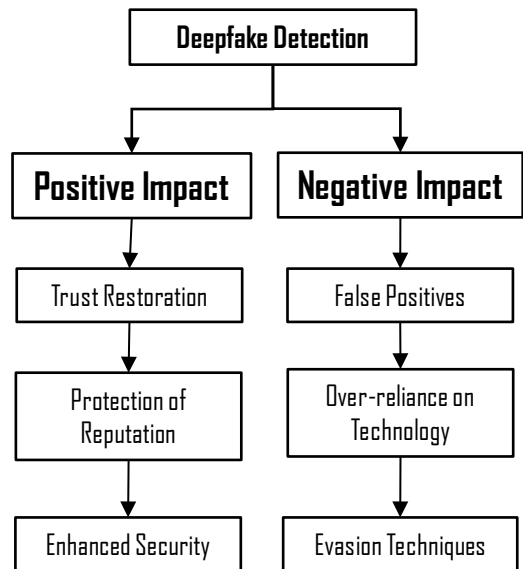




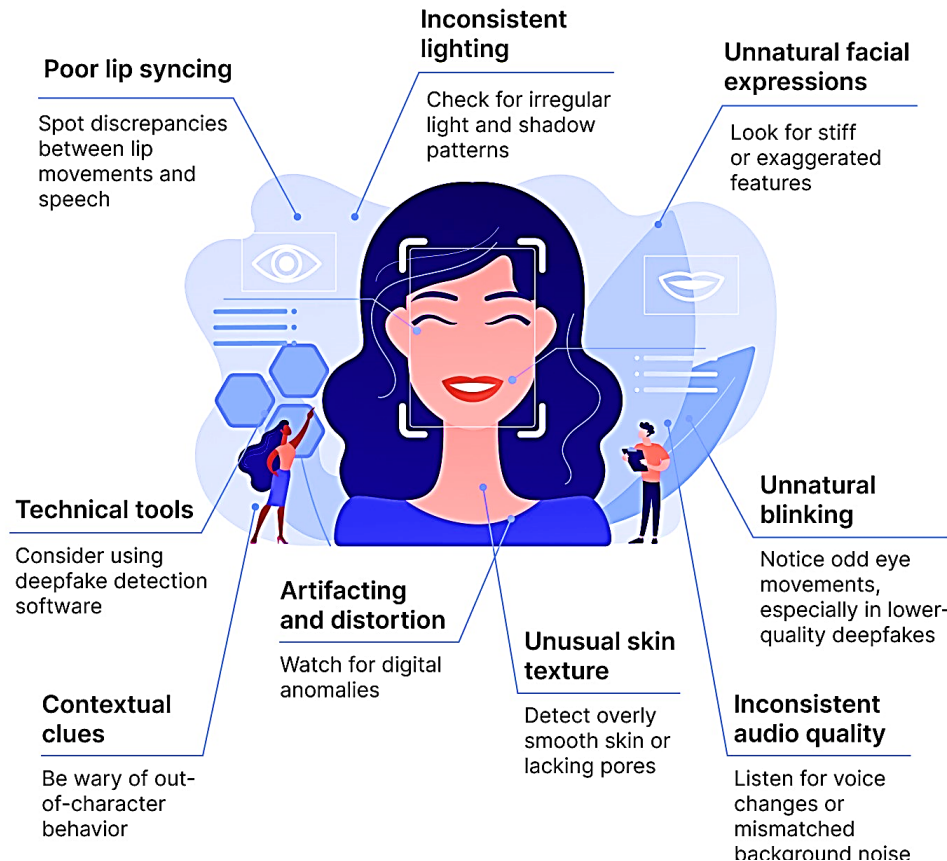
Deep Fake Detection in Action

Description: An image showing a side-by-side comparison of a real video frame and a deep fake video frame, with the detection tool highlighting the differences and marking the fake content.

Purpose: Demonstrates the core functionality of the project—identifying deep fakes.



How to Recognize Deepfake Content



Overall Impact and Benefits:

The solution is poised to create significant positive impacts across various sectors. It addresses critical challenges posed by deep fake technology, offering social, economic, and environmental benefits. By enhancing trust, protecting individuals and organizations, and fostering ethical media practices, the solution will contribute to a safer, more informed, and resilient digital society.

1. Deep Fake Detection Technology

A G-Research on Deepfake Detection

Google research - <https://research.google/blog/contributing-data-to-deepfake-detection-research/>

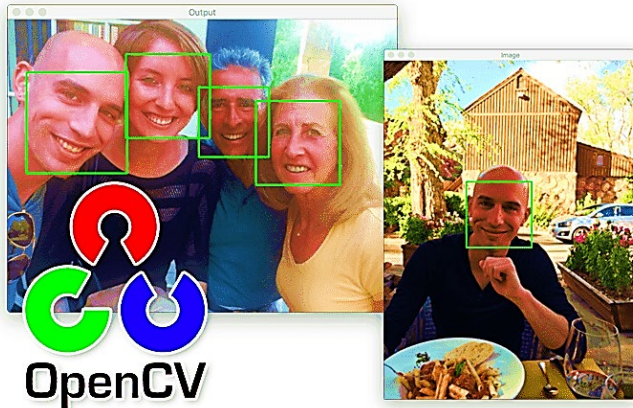
2. Computer Vision and Machine Learning by YouTube

Introduction to OpenCV

YT - <https://www.youtube.com/playlist?list=PLzM8GfZo4-IUA8uGjeXh8UuzPYc6vZRn>

Face Detection using OpenCV

<https://youtu.be/mPCZLOVTEc4?feature=shared>



Open CV Reference

3. Ethics and Legal Implications of Deep Fakes

The Ethics of Deepfakes and Deepfake Detection

Deepfake Journals -

<https://youtu.be/mPCZLOVTEc4?feature=shared>

Legal Implications of Deepfake Technology

Research Gate about Legal Implications -

https://www.researchgate.net/publication/379221500_Deepfakes_The_Legal_Implications

4. Market and Economic Impact

The Economic Impact of Deepfake Technology

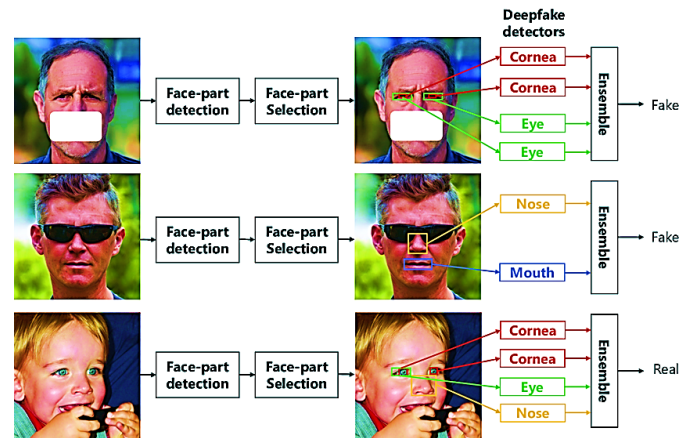
The Economics about Deepfakes -

<https://www.computer.org/csdl/magazine/co/2023/08/10206053/1P1cMAFEiNa>

How Deepfakes Could Impact Global Economies

Openfox about impact of Deepfakes in society -

<https://www.openfox.com/deepfakes-and-their-impact-on-society/>



Deepfake Detection Reference

5. Tools and Frameworks

TensorFlow Documentation

<https://www.tensorflow.org/learn>

YT - <https://www.youtube.com/playlist?list=PLGvjgQ2-qJffU2vF6-IG9DI5a4tR0kzt9>

NumPy Documentation

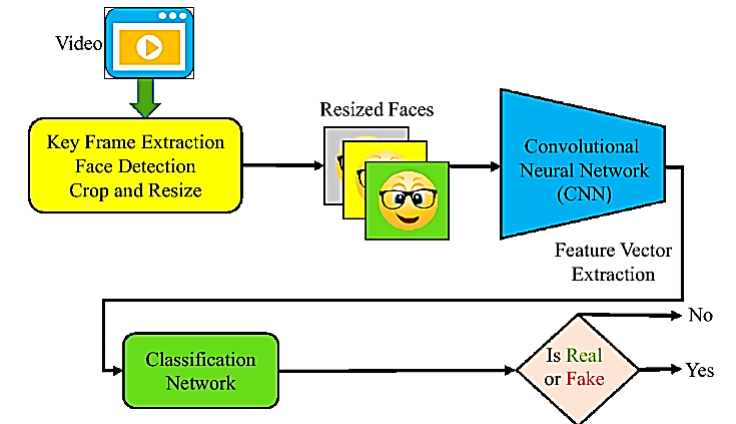
<https://numpy.org/learn/>

YT - https://youtu.be/V_xrolbcAuA?feature=shared

PyTorch Documentation

<https://pytorch.org/tutorials/>

YT - https://www.youtube.com/watch?v=vo_fU0k-IKk&list=PLWKjhJtqVAbm3T2EqI_KgloC7ogdXxdRa



Detection process Reference